



نظم التشغيل (أهلية)

12:00-2:00

22/1/2025

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Faculty of Computers & Information, Assiut University

Course code: CS321

3rd Level

Final Exam

Duration: Two hours

This exam for the following program(s) : (Add the program's name(s))

Points: -/50

1. * الإسم الرباعي (بالعربي فقط)

2. * رقم الجلوس

3. * المستوى

☐ الاول

☐ الثاني

☒ الثالث

☐ الرابع 2014

☐ الرابع 2015

☐ الرابع 2016

☐ الرابع 2017

☐ الرابع 2018

☐ الرابع 2019

☐ الرابع 2020

☐ الرابع 2021

4. البرنامج *

☐ الأهلية علوم حاسب أو نظم معلومات

☐ الأهلية هندسة برمجيات

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Will be reviewed

8. Peterson's solution guarantees:

- ☐ Mutual exclusion
- ☐ Progress
- ☐ Bounded waiting
- ☒ All of the above



Will be reviewed

9. Disk scheduling involves deciding

- ☐ which disk should be accessed next
- ☒ the order in which disk access requests must be serviced
- ☐ the physical location where files should be accessed in the disk
- ☐ none of the above



Will be reviewed

10. A disadvantage of deadlock prevention is that it can be restrictive and may limit system performance. 

☒ True

☐ False

6

 **Will be reviewed**

11. The primary purpose of a TLB is to: 

☐ Reduce the number of page table accesses.

☐ Increase the size of physical memory.

☒ Improve the performance of virtual memory systems.

☐ All of the above.

8

 **Will be reviewed**

12. A deadlock occurs when: 

☒ Two or more processes are waiting for each other to release a resource.

☐ A process is waiting for a resource that is held by another process.

☐ A process is starved of CPU time.

☐ A process is terminated abruptly.

6

 **Will be reviewed**

13. Assume three jobs arrive at approximately the same time, but Job A arrives slightly before Job B, and Job B arrives slightly before job C. Job A requires 400 ms of CPU,

7

Job B is 800 ms, and Job C is 1200 ms. Assume a time-slice of 600 msec. Given a FIFO scheduler, what is the average response time of the three jobs? 

- ☐ 400 ms
- ☒ 533.33 ms
- ☐ 1333.33 ms
- ☐ 1600 ms

 **Will be reviewed**

14. Assume three jobs arrive at approximately the same time, but Job A arrives slightly before Job B, and Job B arrives slightly before job C. Job A requires 400 ms of CPU, Job B is 800 ms, and Job C is 1200 ms. Assume a time-slice of 600 msec. Given a RR scheduler, what is the average response time of the three jobs? 

- ☐ 400 ms
- ☒ 466.67 ms
- ☐ 533.33 ms
- ☐ 1333.33 ms
- ☐ 1600 ms



 **Will be reviewed**

15. Thrashing 

- ☐ reduces page I/O
- ☐ decreases the degree of multiprogramming
- ☒ implies excessive page I/O



- ☐ improves the system performance

✓ **Will be reviewed**

6

16. In the Banker's algorithm, the content of the **matrix Need** is 

- ☐ Allocation – Available
- ☐ Max – Available
- ☒ **Max – Allocation**
- ☐ Allocation – Max

✓ **Will be reviewed**

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17. A disadvantage of the Shortest Job First (SJF) scheduling algorithm is: 

- ☒ **It can lead to starvation of long processes**
- ☐ It is difficult to implement
- ☐ It is not fair
- ☐ It can lead to long wait times for short processes

8

✓ **Will be reviewed**

18. What is a Translation Look-aside Buffer (TLB)? 

- ☒ **A high-speed cache that stores recently used page table entries.**
- ☐ A hardware device that translates physical addresses to virtual addresses.
- ☐ A software algorithm that optimizes memory access.

☐ A type of memory management unit.

📝 **Will be reviewed**

7

19. The page replacement policy that sometimes leads to more page faults when the size of the memory is increased is: 📖

☒ FIFO page replacement

☐ least-recently-used (LRU) page replacement

☐ Optimal page replacement

☐ no such policy exists

📝 **Will be reviewed**

20. A disadvantage of the First-Come-First-Served (FCFS) scheduling algorithm is: 📖

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☒ It can lead to long wait times for short processes

☐ It is difficult to implement

☐ It is not fair

☐ It can lead to starvation

📝 **Will be reviewed**

21. A small computer has 5 page frames. A process makes the following list of page references: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. How many page faults using least-recently-used (LRU) algorithm? 📖

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☐ 7

☒ 8

☐ 10

☐ 12

☐ 14

 **Will be reviewed**

22. The number of virtual pages is always identical to the number of physical pages.



☐ True

☒ False

 **Will be reviewed**

23. Virtual memory is a technique that allows a process to use more memory than is physically available 

☒ True

☐ False

 **Will be reviewed**

24. What is the main goal of disk scheduling algorithms? 

☐ To minimize the number of disk requests.

☐ To maximize disk throughput.

☒ To minimize disk seek time.

☐ To maximize disk utilization.

📝 **Will be reviewed**

25. The `wait()` operation on a semaphore: 📖

☒ Decrements the semaphore value

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☐ Increments the semaphore value

☐ Checks if the semaphore value is positive

☐ None of the above

📝 **Will be reviewed**

26. Which mode of operation provides direct access to system hardware and memory?



☐ User mode

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☒ Kernel mode

☐ Protected mode

☐ Virtual mode

📝 **Will be reviewed**

27. Assume three jobs arrive at approximately the same time, but Job A arrives slightly before Job B, and Job B arrives slightly before job C. Job A requires 400 ms of CPU, Job B is 800 ms, and Job C is 1200 ms. Assume a time-slice of 600 msec. Given a RR scheduler, what is the waiting time of job B? 📖

- ☐ 400 ms
- ☐ 800 ms
- ☒ 1000 ms
- ☐ 1200 ms

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📝 **Will be reviewed**

28. Mutual exclusion is a technique to: 📖

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- ☐ Allow multiple processes to access a shared resource simultaneously.
- ☒ Control access to a shared resource by allowing only one process at a time.
- ☐ Prevent processes from accessing shared resources.
- ☐ Signal the end of a critical section.

📝 **Will be reviewed**

29. Which of the following is an example of synchronous communication? 📖

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- ☐ Email
- ☐ Forum posts
- ☒ Video conferencing
- ☐ Social media posts

📝 **Will be reviewed**

30. A cache coherence protocol is a mechanism that: 📖

- 1
- ☐ Maintains consistency between different caches
 - ☐ Improves cache performance
 - ☐ Reduces cache miss rates
 - ☒ All of the above

📝 **Will be reviewed**

31. A race condition occurs when: 📖

- ☒ Two or more processes access a shared resource concurrently.
 - ☐ A process is terminated abruptly.
 - ☐ A process is waiting for a resource that is held by another process.
 - ☐ A process is starved of CPU time.
- 4

📝 **Will be reviewed**

32. Which of the following scenarios can lead to priority inversion? 📖

- ☒ A high-priority task waiting for a resource held by a low-priority task.
 - ☐ A low-priority task waiting for a resource held by a high-priority task.
 - ☐ Two high-priority tasks competing for the same resource.
 - ☐ Two low-priority tasks competing for the same resource.
- 5

📝 **Will be reviewed**

33. If there are 32 pages, each of size 1 Kbytes, then the logical address should have



☐ 10 bits

☐ 14 bits

☒ 15 bits

☐ 16 bits

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Will be reviewed

34. Which memory allocation technique divides physical memory into fixed-size blocks called frames?

☐ Contiguous memory allocation

☒ Paging

☐ Segmentation

☐ Virtual memory

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Will be reviewed

35. A small computer has 5 page frames. A process makes the following list of page references: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. How many page faults using optimal page replacement algorithm?

☒ 7

☐ 8

☐ 10

☐ 12

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☐ 14

✓ **Will be reviewed**

36. A test-and-set instruction is an atomic operation that: 

- ☐ Reads a value from memory
- ☐ Sets a value in memory
- ☒ Reads a value from memory, sets it to a new value, and returns the old value
- ☐ None of the above

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✓ **Will be reviewed**

37. Which scheduling algorithm assigns a fixed amount of time to each process? 

- ☐ First-Come-First-Served (FCFS)
- ☐ Shortest Job First (SJF)
- ☐ Priority Scheduling
- ☒ Round Robin

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✓ **Will be reviewed**

38. What is external fragmentation? 

- ☐ When memory is allocated in fixed-size blocks and a process requires a block larger than any available block.
- ☐ When memory is allocated in variable-size blocks and there is unused space within allocated blocks.

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- ☒ When memory is fragmented into small, non-contiguous blocks that are too small to satisfy a memory request.
- ☐ When a process is allocated more memory than it needs.

 Will be reviewed

39. Assume three jobs arrive at approximately the same time, but Job A arrives slightly before Job B, and Job B arrives slightly before job C. Job A requires 400 ms of CPU, Job B is 800 ms, and Job C is 1200 ms. Assume a time-slice of 600 msec. Given a RR scheduler, what is the turnaround time of job B? 

- ☐ 800 ms
- ☐ 1200 ms
- ☐ 1600 ms
- ☒ 1800 ms
- ☐ 2400 ms



 Will be reviewed

40. A spinlock is a type of lock that: 

- ☐ Blocks the process until the lock is released
- ☒ Continuously checks the lock until it becomes available
- ☐ Uses a timer to release the lock after a certain period
- ☐ None of the above



✓ **Will be reviewed**

41. The Producer-Consumer Problem involves: 

- ☒ A single producer and a single consumer
- ☐ Multiple producers and a single consumer
- ☐ A single producer and multiple consumers
- ☐ Multiple producers and multiple consumers

3

✓ **Will be reviewed**

42. In non-preemptive scheduling, a running process: 

- ☐ Can be interrupted by the OS
- ☒ Cannot be interrupted until it completes or voluntarily releases the CPU
- ☐ Is always interrupted after a fixed time quantum
- ☐ Is always interrupted when a higher priority process arrives

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✓ **Will be reviewed**

43. Which disk scheduling algorithm can lead to starvation, as it may not service requests that are far away from the current head position? 

- ☐ FCFS
- ☒ SSTF
- ☐ SCAN

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☐ LOOK

📝 **Will be reviewed**

44. Which of the following is NOT a common scheduling criterion? 📖

☐ CPU utilization

☐ Throughput

☐ Response time

☒ Memory usage

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📝 **Will be reviewed**

45. An I/O-bound process is a process that: 📖

☒ Spends more time waiting for I/O operations than executing CPU instructions.

☐ Spends more time executing CPU instructions than waiting for I/O operations.

☐ Spends equal amounts of time on I/O and CPU operations.

☐ Does not perform any I/O operations.

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📝 **Will be reviewed**

46. What is a page fault? 📖

☐ A hardware error that occurs when a memory access fails.

☐ A software error that occurs when a program tries to access invalid memory.

☒ An event that occurs when a requested page is not present in physical memory.

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- ☐ A technique used to allocate memory to processes.

📝 **Will be reviewed**

47. The best-fit algorithm can be used for: 📝

- ☒ contiguous allocation
- ☐ linked allocation
- ☐ indexed allocation
- ☐ FAT allocation

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📝 **Will be reviewed**

48. Assume three jobs arrive at approximately the same time, but Job A arrives slightly before Job B, and Job B arrives slightly before job C. Job A requires 400 ms of CPU, Job B is 800 ms, and Job C is 1200 ms. Assume a time-slice of 600 msec. Given a RR scheduler, what is the waiting time of job C? 📝

- ☐ 400 ms
- ☐ 800 ms
- ☒ 1200 ms
- ☐ 2400 ms

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📝 **Will be reviewed**

49. A critical section is a piece of code that: 📝

- ☒ Must be executed by only one process at a time.

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- ☐ Can be executed by multiple processes simultaneously.
- ☐ Is not essential for correct program execution.
- ☐ Is executed only once during program execution.

📝 **Will be reviewed**

50. In a page memory, the page hit ratio is 0.9. The time required to access a page in secondary memory is equal to 100ns. The time required to access a page in primary memory is 10ns. The average time required to access a page is 📎

- ☐ 9 ns
- ☐ 10 ns
- ☒ 19 ns
- ☐ 90 ns

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📝 **Will be reviewed**

51. A system call is a request from a user program to the operating system kernel. When a system call is made, the CPU typically switches from: 📎

- ☐ Kernel mode to user mode
- ☒ User mode to kernel mode
- ☐ Protected mode to user mode
- ☐ Virtual mode to kernel mode

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📝 **Will be reviewed**

52. Which of the following is NOT an Interprocess Communication (IPC) mechanism?



- ☐ Shared Memory
- ☐ Message Passing
- ☒ Threads

3

Will be reviewed

53. Peterson's solution is limited to:

- ☐ Two processes
- ☒ Any number of processes
- ☐ A fixed number of processes
- ☐ None of the above

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Will be reviewed

54. A small computer has 5 page frames. A process makes the following list of page references: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. How many page faults using FIFO algorithm?

- ☐ 7
- ☐ 8
- ☒ 10
- ☐ 12

7

☑ **Will be reviewed**

55. When a user process needs to perform an I/O operation, it typically: 

- ☐ Directly accesses the I/O device.
- ☒ Executes a system call to request the kernel to perform the operation.
- ☐ Switches to kernel mode and performs the operation itself.
- ☐ Pauses execution until the I/O device becomes available.

↑

☑ **Will be reviewed**

56. Which disk scheduling algorithm prioritizes the request closest to the current head position? 

- ☐ First-Come-First-Served (FCFS)
- ☒ Shortest Seek Time First (SSTF)
- ☐ SCAN
- ☐ LOOK

9

☑ **Will be reviewed**

57. Which method involves blocking one of the four necessary conditions for a deadlock? 

- ☒ Deadlock prevention
- ☐ Deadlock avoidance
- ☐ Deadlock detection and recovery

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☐ None of the above



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